

This notebook (model_Performance_Measurement.ipynb file) demonstrates how to evaluate machine learning model performance using key metrics such as accuracy, F1 score, R^2 , and the Area Under the Curve (AUC). It shows how different evaluation measures capture different aspects of model quality, depending on whether the task is classification or regression.

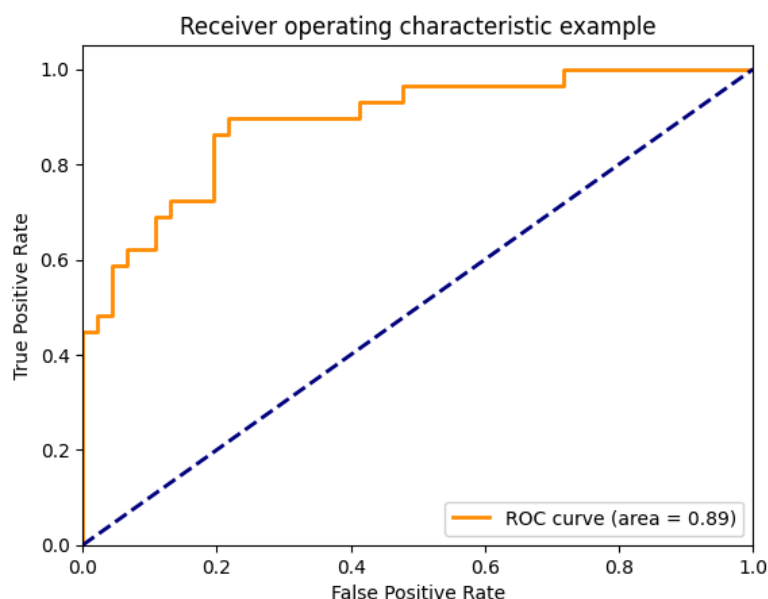
The notebook explores how model performance changes when parameters are adjusted, highlighting that good performance on one metric does not necessarily imply overall model quality. For example, accuracy alone may be misleading in classification problems, while metrics such as AUC provide a more robust view of model discrimination ability. Similarly, R^2 is used to assess how well a regression model explains the variance in the data.

The F1 score is also a very common performance metric used in classification tasks that balances precision and recall. Precision measures how many of the predicted positive cases are actually correct, while recall measures how many of the actual positive cases are correctly identified. The F1 score is the harmonic mean of these two metrics, meaning it gives a high value only when both precision and recall are high. This makes it particularly useful in situations where the data is imbalanced, or where both false positives and false negatives are important. Unlike accuracy, which can be misleading in such cases, the F1 score provides a more reliable measure of a model's effectiveness.

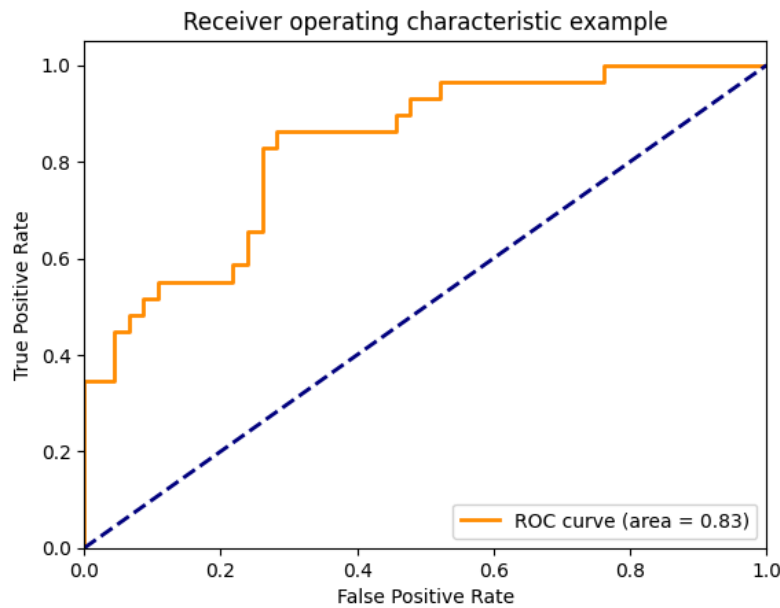
Overall, the exercise demonstrates that model evaluation is a critical step in machine learning, and that selecting appropriate performance metrics is essential for correctly interpreting model behaviour.

For example In case of the support vector classifier the polynomial kernel gave better results than the linear for AUC for each random state I tried:

“poly”



“linear”



In our publication we used the G-score as a metric

(<https://www.nature.com/articles/s41598-025-32123-6> see also under ‘my publications’).

The G-score is a composite performance metric that combines clinical accuracy, bias, and fairness into a single measure ranging between 0 and 1, where higher values indicate better model performance. Unlike standard metrics, it incorporates a clinically meaningful error threshold by treating prediction errors within $\pm\frac{1}{2}$ MCID (minimally clinically important difference) as negligible. The metric is based on three components: relative clinical accuracy (RCA), relative clinical bias (RCB), and fairness (F), all scaled to the [0,1] interval. Fairness is quantified using the Predictive Power Score (PPS), which captures both linear and non-linear relationships between prediction errors and protected variables; the fairness component is defined as one minus this association. The overall G-score is calculated as a weighted combination of accuracy and bias (via the product $RCA \times RCB$) and fairness, allowing explicit control over the importance of fairness through a weighting parameter. This makes the G-score particularly useful in applied settings where predictive performance must be evaluated alongside fairness considerations.